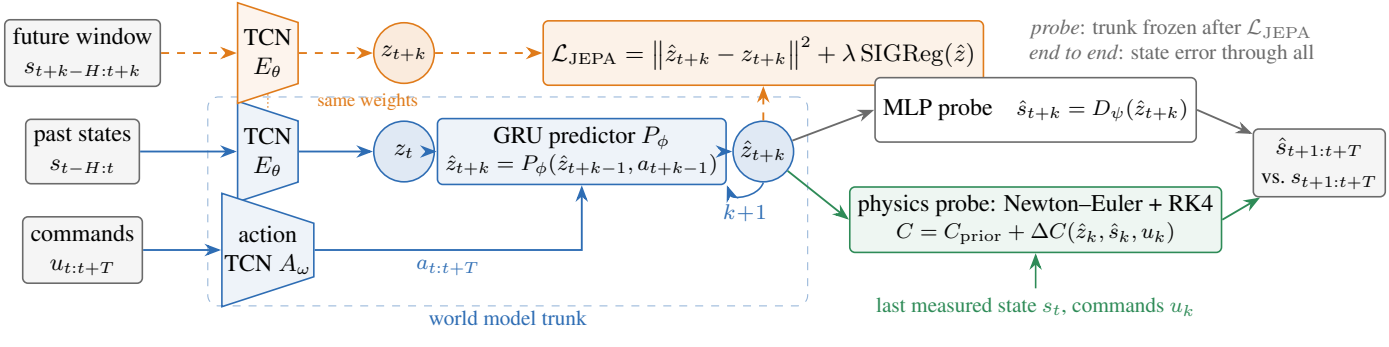
future window
$s_{t+k-H:t+k}$
TCN
E_θ
z_{t+k}
$\mathcal{L}_{\text{JEPA}} = \|\hat{z}_{t+k} - z_{t+k}\|^2 + \lambda\,\text{SIGReg}(\hat{z})$
probe: trunk frozen after $\mathcal{L}_{\text{JEPA}}$
end to end: state error through all
same weights
past states
$s_{t-H:t}$
TCN
E_θ
z_t
GRU predictor P_ϕ
$\hat{z}_{t+k} = P_\phi(\hat{z}_{t+k-1}, a_{t+k-1})$
$\hat{z}_{t+k}$
$k{+}1$
MLP probe $\hat{s}_{t+k} = D_\psi(\hat{z}_{t+k})$
$\hat{s}_{t+1:t+T}$
vs. $s_{t+1:t+T}$
commands
$u_{t:t+T}$
action
TCN A_ω
$a_{t:t+T}$
physics probe: Newton–Euler + RK4
$C = C_{\text{prior}} + \Delta C(\hat{z}_k, \hat{s}_k, u_k)$
world model trunk
last measured state s_t, commands u_k